RAG-Based Customer Support Assistant (with LangGraph + HITL)

LOW-LEVEL DESIGN

1. MODULE-LEVEL DESIGN

The system is divided into the following modules:

- Document Processing Module
  - Loads the PDF file
  - Extracts text from document

- Chunking Module
  - Splits large text into smaller chunks
  - Improves retrieval accuracy

- Embedding Module
  - Converts text chunks into vectors
  - Uses HuggingFace embedding model

- Vector Storage Module
  - Stores embeddings in ChromaDB
  - Supports similarity search

- Retrieval Module
  - Takes user query
  - Finds most relevant chunks

🔹 Query Processing Module

- Processes user input
- Prepares query for retrieval

🔹 Graph Execution Module (LangGraph)

- Controls workflow execution
- Connects different modules

🔹 HITL Module (Human-in-the-Loop)

- Handles cases where system fails
- Sends query to human support

🟢 2. DATA STRUCTURES

🔹 Document Structure

```
{
 "page_content": "text from PDF",
 "metadata": {}
}
```

🔹 Chunk Structure

```
{
 "chunk_id": 1,
 "text": "small part of document"
}
```

🔹 Embedding Structure

```
[0.12, 0.98, 0.45, ...]
```

- Query-Response Structure

```
{
 "query": "What is refund policy?",
 "answer": "Customers can request refund within 7 days"
}
```

- State Object (LangGraph)

```
{
 "query": "",
 "retrieved_chunks": [],
 "answer": "",
 "status": "AI or HUMAN"
}
```

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3. WORKFLOW DESIGN (LangGraph)

The workflow consists of:

- Nodes
  - Input Node
  - Processing Node
  - Output Node

- Flow

Input → Retrieve → Generate Answer → Output

- State Flow
  - Query enters system
  - Chunks retrieved
  - Answer generated
  - Passed to output

🟢 4. CONDITIONAL ROUTING LOGIC

The system decides:

- If relevant chunks found → generate answer
- If no chunks found → escalate to human
- If query is complex → escalate
- If low confidence → escalate

🟢 5. HITL DESIGN

🔹 When triggered:

- No relevant answer
- Low confidence

🔹 Process:

1. Query sent to human
2. Human generates answer
3. Response returned to user

🟢 6. API / INTERFACE DESIGN

🔹 Input Format

```
{
 "query": "What is refund policy?"
}
```

🔹 Output Format

```
{
 "answer": "Refund within 7 days",
 "status": "AI"
}
```

🟢 7. ERROR HANDLING

The system handles:

- Missing PDF file
- No chunks generated
- No relevant results
- Model failure

Fallback:

- Show message
- Trigger HITL